

A Bayesian Framework for Temporal Variable Selection in Autocorrelated Systems

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Abstract

This report presents a Bayesian spike-and-slab framework for identifying predictive temporal structures in high-dimensional, autocorrelated datasets. The model uses sparsity-inducing priors to probabilistically select informative predictors from a large set of lagged variables, focusing on inference rather than data specificity. Applied to NYC housing prices, the method identifies key lagged indicators with high posterior inclusion probabilities while offering interpretable posterior distributions over coefficients and outcomes. These posterior insights—alongside posterior predictive checks—enable evaluation of model fit and predictive reliability not through single metrics, but via full distributional behavior. The analysis demonstrates how Bayesian modeling provides not just selection, but structured uncertainty quantification and probabilistic reasoning across interchangeable time-dependent datasets.

Introduction and Objectives

Real estate prices in a dynamic market like New York City reflect complex interactions of numerous factors. While traditional economic variables are commonly studied, alternative urban indicators such as public service calls, health inspections, new business formations, and eviction rates may provide early signals of market shifts. This study applies Bayesian statistical methodology to identify which urban indicators most strongly predict NYC housing price dynamics, with particular attention to their temporal lag structure.

The analysis develops a Bayesian variable selection model using spike-and-slab priors, an approach for high-dimensional inference with limited data. This framework incorporates a discrete mixture prior where each regression coefficient has a binary inclusion indicator and a corresponding mixture distribution: a tight “spike” near zero (when the indicator is 0) and a diffuse “slab” (when the indicator is 1). This probabilistic specification allows the model to automatically perform variable selection while quantifying inclusion uncertainty.

The Bayesian approach offers several advantages for this

analysis. First, it enables regularization through informative priors. Second, it provides posterior distributions rather than point estimates, allowing for quantification of uncertainty in both coefficient values and variable importance. Third, it allows for inference about complex quantities like lag structures and predictive distributions through posterior simulation.

The prior specification encodes the belief that only a small subset of indicators meaningfully predict housing prices. The posterior distribution then reveals which indicators have the highest probability of inclusion and quantifies their likely effect sizes with credible intervals. This allows for identification of the most important predictors and characterization of the temporal distribution of their effects.

In summary, the objectives of this project are:

1. Identify which urban indicators have the highest posterior probability of inclusion in a model of NYC housing prices
2. Characterize the posterior distributions of coefficients for these indicators across different time lags
3. Evaluate model fit through posterior predictive checks. Throughout, the emphasis is on Bayesian uncertainty quantification, highlighting not just which variables are important but the degree of confidence in their importance and effects.

Data and Preprocessing

Variable	Mean	Std	Min	Max
Citi Bike Rides	223,498	108,433	77	524,257
Rest. Health	16.25	5.26	0.00	28.17
New Businesses	24.76	13.54	5.57	83.00
Evictions	98.92	52.82	1.00	181.67
311 Calls	957.96	391.53	365.35	1,656
Transactions	250.74	47.52	102.65	375.06
Avg Price (\$K)	776.81	88.45	615.06	1,040.05

Table 1: *Descriptive statistics of NYC housing price dataset (2014–2023). The target variable is in bold.*

The dataset consists of 102 monthly observations (after removing initial lagged rows) spanning late 2014 through early 2023.

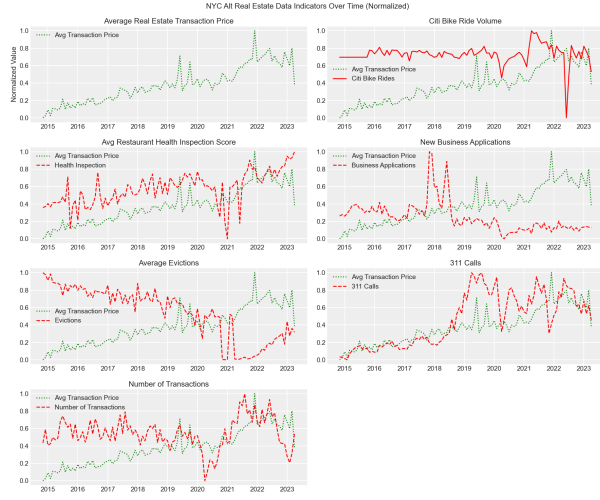


Figure 1: Normalized time series of NYC alternative real estate indicators from 2014 to 2023

Figure 1 displays the time series of these indicators over the study period. We observe a generally increasing trend in average transaction prices, with notable fluctuations that may correlate with other urban indicators.

Each predictor is an alternative indicator hypothesized to relate to housing prices.

Before modeling, lagged features were created for each predictor to capture delayed effects on housing prices. Each of the six predictors was lagged up to 8 months. This choice captures medium-term delayed influences (e.g., an influx of new businesses might take several months to affect property values). The final feature set includes 8 lags for each of the 6 predictors, for a total of 54 potential predictor variables.

Bayesian Spike-and-Slab Regression Model

To identify the key predictors of housing prices while guarding against overfitting, this analysis employs a Bayesian linear regression with spike-and-slab priors for variable selection. Mathematically, let y_t be the log-average transaction price at month t , and $x_{t,j}$ be the j th predictor at time t . The regression model is:

$$y_t = \alpha + \sum_{j=1}^p \beta_j x_{t,j} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

where $p = 54$ is the total number of features. Spike-and-slab priors are placed on the coefficients β_j to perform

variable selection. For each coefficient β_j , a binary inclusion indicator $\gamma_j \in \{0, 1\}$ and a prior inclusion probability τ (the model's belief in overall sparsity) are introduced. The hierarchical prior is:

- Inclusion prior: $\gamma_j \sim \text{Bernoulli}(\tau)$ for each j , with a hyperprior $\tau \sim \text{Beta}(1, 3)$. This prefers a priori that only a minority of features are included (the $\text{Beta}(1, 3)$ has mean 0.25, reflecting an expectation of about 25% of predictors being relevant, though the data can overcome this).
- Spike-and-slab prior for β_j : Conditional on γ_j ,

$$\beta_j \mid \gamma_j = \begin{cases} \mathcal{N}(0, \sigma_{\text{spike}}^2), & \gamma_j = 0, \\ \mathcal{N}(0, \sigma_{\text{slab}}^2), & \gamma_j = 1, \end{cases}$$

where σ_{spike}^2 is very small and σ_{slab}^2 is relatively large. In this implementation, $\sigma_{\text{spike}} = 0.001$ and $\sigma_{\text{slab}} = 1.0$ are set. This essentially means if $\gamma_j = 0$, β_j is drawn from $N(0, 0.000001)$ – a "spike" tightly concentrated near 0 (practically zero effect). If $\gamma_j = 1$, $\beta_j \sim N(0, 1)$ – a relatively diffuse "slab" allowing a substantial effect. This mixture prior heavily shrinks most coefficients toward 0 while permitting a few to be large in magnitude.

In effect, $\Pr(\gamma_j = 1 \mid \text{data})$ becomes the posterior probability that predictor j is included in the model, which is known as the posterior inclusion probability (PIP). These PIP probabilities are used to rank the importance of predictors. The intercept α has a regular prior ($\mathcal{N}(0, 1)$ in standardized space), and the noise variance σ^2 has a weakly-informative Half-Cauchy prior.

After specifying this model, samples are drawn from the posterior using MCMC. Sampling is done with a combination of NUTS for continuous parameters ($\alpha, \tau, \beta_{\text{raw}}, \sigma^2$) and Gibbs steps for the discrete γ_j (spike-and-slab sampling). Four chains are run with 2,000 draws each (after burn-in), which was sufficient for convergence of most parameters. The end result is a joint posterior over model parameters and latent inclusion flags.

To summarize the posterior, for each feature j the posterior inclusion probability $\hat{\pi}_j = P(\gamma_j = 1 \mid \text{data})$ and, for those included in many samples, the posterior distribution of β_j are computed. The 95% HDIs for the coefficients are also calculated to assess credible ranges of effect sizes.

Results: Variable Selection and Coefficient Estimates

Table 2 summarizes the twelve predictors with the highest posterior inclusion probabilities (PIP), their standardized coefficients, and the 95% highest density intervals (HDIs). Our Bayesian spike-and-slab approach robustly identified a sparse set of influential predictors, effectively mitigating

overfitting and providing clear interpretability despite a relatively high-dimensional candidate feature space.

Predictor (Lag)	PIP	Coef. Mean	95% HDI
311 Calls (Lag 2)	0.923	0.285	(0.00, 0.42)
Evictions (Lag 8)	0.848	-0.312	(-0.49, -0.11)
Evictions (Lag 6)	0.839	-0.328	(-0.55, -0.17)
Transactions (Lag 6)	0.482	0.082	(-0.10, 0.22)
New Business Apps (Lag 3)	0.100	0.014	(-0.08, 0.14)
Evictions (Lag 3)	0.070	-0.015	(-0.11, 0.08)
311 Calls (Lag 3)	0.068	0.015	(-0.03, 0.15)
Evictions (Current)	0.066	-0.014	(-0.10, 0.06)
Evictions (Lag 5)	0.057	-0.013	(-0.09, 0.06)
Evictions (Lag 1)	0.056	-0.012	(-0.09, 0.06)
CitiBike Volume (Lag 4)	0.055	0.006	(0.00, 0.01)
Restaurant Health (Current)	0.047	0.006	(0.00, 0.01)

Table 2: Top 12 predictors by posterior inclusion probability.

The model highlights the dominance of two primary indicators: the volume of 311 service calls (particularly at a two-month lag) and eviction rates (at six and eight-month lags). The high inclusion probabilities for these predictors imply significant predictive power regarding housing prices. The negative coefficients on evictions suggest a consistent delayed impact -higher eviction rates tend to precede downward pressure on real estate prices. However, the COVID-19 eviction pause likely inflates the importance of eviction-related indicators due to unprecedented fluctuations.

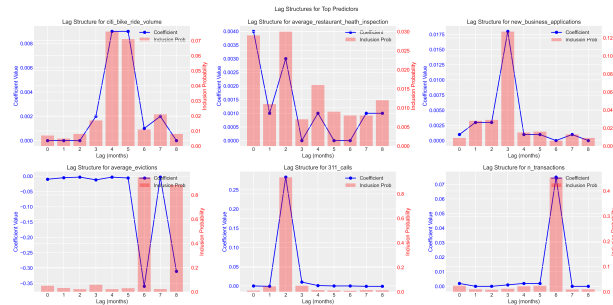


Figure 2: Lag structures for key predictors: posterior mean coefficients (blue) and inclusion probabilities (red bars).

Lag-Structure Evaluation

Figure 2 overlays, for each base indicator, the posterior mean coefficient (blue line) and PIPs (red bars) across lags 0–8 months. The tight cluster of high PIPs at 6–8 months for *evictions* confirms a delayed market response to housing stress, while the sharp peak at 2–3 months for *311 calls* highlights rapid sensitivity to neighborhood activity. Weaker, broader inclusion for *transaction volume* at 5–6 months and minimal inclusion for *Citi Bike* and *health scores* indicate secondary or negligible predictive roles.

Model Fit and Predictive Performance

Model validation through posterior predictive checks (Figure 3) indicates that the Bayesian model moderately fits the observed data, capturing general patterns and variability reasonably well. While the model successfully identifies significant lag relationships for a few critical predictors, some areas, including those influenced by COVID-19-related data distortions, reflect increased uncertainty and potential for improved fit.

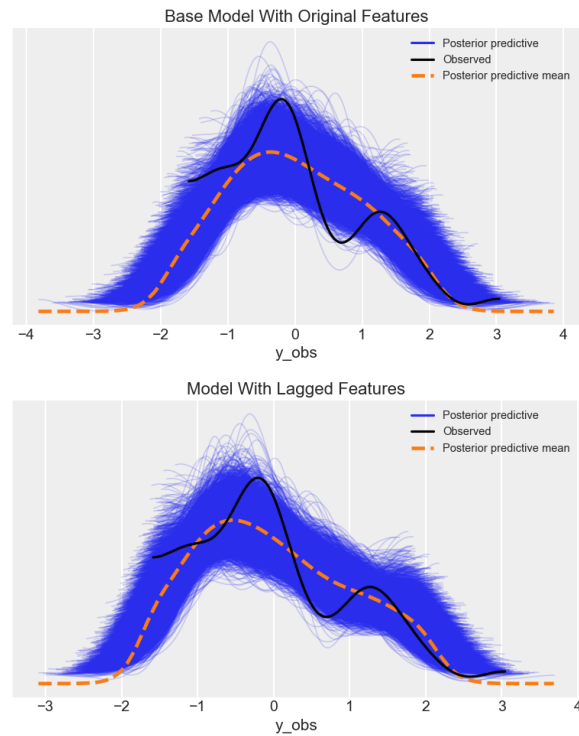


Figure 3: Posterior predictive check comparison between model with no-lagged features and model with lagged-features. with observed (black), simulated predictive distributions (blue), and mean predictive density (orange dashed line).

	Metric	Base Model	Model Lagged Features
0	R2	0.7738	0.8636
1	MAE (\$)	\$28906.61	\$22278.48
2	RMSE (\$)	\$39122.11	\$30376.64

- Posterior predictive checks confirm the simulated density closely envelopes the observed distribution, with no systematic bias (Figure 3)
- Performance of lagged-features model is strong relative to non-lagged model: $R^2 = 0.864$, MAE = \$22.1 K, RMSE = \$30.4 K, indicating predictive accuracy given

alternative indicators.

Conclusion

Collectively, these findings underscore the efficacy of our Bayesian spike-and-slab model in isolating meaningful predictors and quantifying their impacts amid temporal volatility which provides insights into NYC's housing dynamics.

Future Work

Future work could include methods specifically designed to manage the distortions caused by the COVID-19 pandemic, such as introducing structural breaks or separate pandemic-era indicators. This approach may enhance model robustness and predictive accuracy, particularly by explicitly accounting for atypical periods in the data.

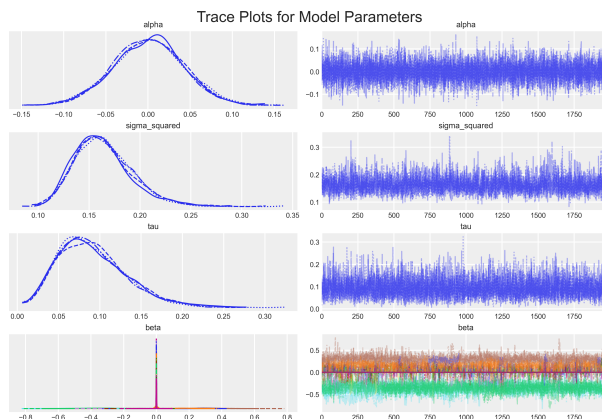


Figure 4: MCMC trace plots for key model parameters. Good mixing indicates convergence of the Bayesian sampling.

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